

Solving problems with the help of systems

Two unknowns, two sentences, two equations — and an answer that names both quantities.

LESSON 150–152

3 × 40 MIN

MATEMATIKA 7, §64

BEYOND STAGE 8

LESSON CLOCK

10

30'

35'

30'

15'

When two unknowns are needed	10 min
Money and quantities	30 min
Numbers and digits	35 min
Geometry and motion	30 min
Checking and rejecting	15 min

LEARNING OBJECTIVES

- Choose two unknowns and say in words what each stands for.
- Turn two conditions into two equations.
- Solve the system by the method that suits it.
- Check both numbers against the original words and reject impossible answers.

TEXTBOOK

National	pp. 451–461
Cambridge	Stage 8 extension
Workbook	Extension sheet 2F

01

Explanation

When two unknowns are needed

If the problem names two quantities and gives two facts relating them, two letters and two equations are usually shorter than one letter and a long expression.

STEP	WHAT TO DO
1	let x and y be the two quantities, and say which is which
2	write the first condition as an equation
3	write the second condition as a second equation
4	solve the system
5	answer both parts of the question, with units

TWO FACTS, TWO EQUATIONS

Count the conditions in the problem before starting. Two unknowns need two independent facts; if only one is given, the problem has infinitely many answers and something has been misread.

Money and quantities

PROBLEM	UNKNOWNNS	SYSTEM	ANSWER
3 pens and 2 pencils cost 13 000; 1 pen and 1 pencil cost 5000	prices x and y	$3x + 2y = 13000, x + y = 5000$	pen 3000, pencil 2000
20 coins of 100 and 500 sum, worth 4400 in all	counts x and y	$x + y = 20, 100x + 500y = 4400$	14
a class of 30 with 4 more girls than boys	counts x and y	$x + y = 30, y - x = 4$	13

ONE EQUATION COUNTS, THE OTHER VALUES

In coin and ticket problems the first equation counts the items and the second adds up their worth. Recognising that pattern writes both equations almost automatically.

Numbers and digits

A two-digit number with tens digit a and units digit b is $10a + b$ — not ab , which would be a product.

STATEMENT	AS ALGEBRA
the number	$10a + b$
the sum of its digits	$a + b$
the number with its digits reversed	$10b + a$

Example: the digits of a two-digit number add to 9, and reversing them gives a number 27 larger.

STEP	WORKING
first condition	$a + b = 9$
second condition	$10b + a = 10a + b + 27$
simplify it	$9b - 9a = 27, \text{ so } b - a = 3$
solve	$a = 3, b = 6$
the number	36

DIGITS ARE WHOLE AND THE TENS DIGIT IS NOT ZERO

A root of $a = 4.5$ or $a = 0$ must be rejected: no such two-digit number exists. The algebra does not know this — the reader does.

Geometry and motion

PROBLEM	SYSTEM	ANSWER
a rectangle of perimeter 36 whose length exceeds its width by 4	$2(x + y) = 36, y - x = 4$	7 by 11
two angles are complementary and differ by 20°	$x + y = 90, y - x = 20$	35° and 55°
a boat goes 20 km/h downstream and 12 km/h up	$x + y = 20, x - y = 12$	boat 16, current 4

DOWNSTREAM AND UPSTREAM

The current adds its speed going down and subtracts it coming up, so the two speeds are $x + y$ and $x - y$. Adding the equations gives the boat's own speed; subtracting gives the current's.

Checking and rejecting

ANSWER	QUANTITY	VERDICT
$(3000; 2000)$	two prices	accepted
$(14; 6)$	two counts of coins	accepted — both whole
$(4.5; 4.5)$	two counts of pupils	rejected — not whole
$(-2; 8)$	two lengths	rejected — negative

CHECK IN THE WORDS, NOT IN THE EQUATIONS

Substituting into your own equations only tests your arithmetic. Reading the answer back into the original sentences also tests the translation, which is where the marks are.

02

Worked examples

EXAMPLE 1 3 pens and 2 pencils cost 13 000 sum; a pen and a pencil together cost 5000. Find each price.

- 1 Let x be the price of a pen and y of a pencil.
- 2 $3x + 2y = 13\ 000$ and $x + y = 5000$.
- 3 From the second, $y = 5000 - x$; substitute: $3x + 10\ 000 - 2x = 13\ 000$.
- 4 $x = 3000$, so $y = 2000$. Check: $9000 + 4000 = 13\ 000$ ✓

ANSWER

Pen 3000, pencil 2000 sum

EXAMPLE 2 The digits of a two-digit number add to 9, and reversing them gives a number 27 larger. Find it.

- 1 Let the digits be a (tens) and b (units).
- 2 $a + b = 9$
- 3 $10b + a = 10a + b + 27$, which is $b - a = 3$.
- 4 $a = 3, b = 6$ — the number is 36, and $63 - 36 = 27$ ✓

ANSWER

36

EXAMPLE 3 A boat travels at 20 km/h downstream and 12 km/h upstream. Find the speed of the boat and of the current.

- 1 Let x be the boat's speed and y the current's.
- 2 $x + y = 20$ and $x - y = 12$.
- 3 Add: $2x = 32$, so $x = 16$.
- 4 $y = 4$. Check: $16 + 4 = 20$ ✓ and $16 - 4 = 12$ ✓

ANSWER

Boat 16 km/h, current 4 km/h

03 Interactive model

Ask the class to write only the two equations for six problems and collect them before any solving; almost every lost mark in this block is a translation error, not an algebraic one.

Write the two equations

A pen x and a pencil y cost 5000 together:

$$x + y = 5000$$

$$xy = 5000$$

$$x - y = 5000$$

$$x = 5000y$$

Three pens and two pencils cost 13 000:

$$3x + 2y = 13000$$

$$5xy = 13000$$

$$3x = 2y$$

$$x + y = 13000$$

20 coins altogether:

$$x + y = 20$$

$$xy = 20$$

$$100x + 500y = 20$$

$$x - y = 20$$

Their value is 4400 sum:

$$x + y = 4400$$

$$100x + 500y = 4400$$

$$600xy = 4400$$

$$x = 4400$$

A two-digit number with digits a and b is:

$$ab$$

$$a + b$$

$$10a + b$$

$$10b + a$$

Reversed, it is:

$$10a + b$$

$$10b + a$$

$$a + b$$

$$ba$$

A rectangle of perimeter 36:

$$x + y = 36$$

$$2(x + y) = 36$$

$$xy = 36$$

$$x - y = 36$$

Downstream speed with current y :

$$x - y$$

$$x + y$$

$$xy$$

$$x$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Two unknowns	Ikki noma'lum	Два неизвестных
Condition	Shart	Условие
To denote	Belgilamoq	Обозначить
Total cost	Umumiy narx	Общая стоимость
Digit	Raḡam	Цифра
Two-digit number	Ikki xonali son	Двузначное число
Perimeter	Perimetr	Периметр
To interpret	Talqin qilmoq	Истолковать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Two unknowns need how many independent facts?

A two-digit number with digits a , b is:

Pens x and pencils y : $3x + 2y = 13\,000$ and $x + y = 5000$ give a pen at:

Complementary angles differing by 20° are:

30° and 50°

35° and 55°

40° and 60°

45° and 65°

Downstream 20, upstream 12: the current is:

4

8

16

32

An answer of 4.5 pupils is:

accepted

rejected

rounded up

doubled

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Two numbers add to 12 and differ by 4

ANSWER 8 and 4

2. A class of 30 with 4 more girls than boys

ANSWER 17 girls, 13 boys

3. Complementary angles differing by 20°

ANSWER 55° and 35°

4. A two-digit number with digits a, b

ANSWER $10a + b$

5. The same number reversed

ANSWER $10b + a$

6. Downstream speed with current y

ANSWER $x + y$

7. Upstream speed

ANSWER $x - y$

Medium · the standard question

7 PROBLEMS

1. 3 pens and 2 pencils cost 13 000; one of each costs 5000

ANSWER Pen 3000, pencil 2000

2. 20 coins of 100 and 500 worth 4400

ANSWER 14 of 100, 6 of 500

3. A rectangle of perimeter 36, length 4 more than width

ANSWER 7 by 11

4. Downstream 20, upstream 12

ANSWER Boat 16, current 4

5. Digits add to 9, reversal is 27 larger

ANSWER 36

6. Two numbers: three times the first plus the second is 11, and they add to 5

ANSWER (3; 2)

7. Supplementary angles differing by 40°

ANSWER 110° and 70°

Hard · stretch and reason

7 PROBLEMS

1. Digits add to 11 and the reversal is 27 smaller

ANSWER 74

2. 5 apples and 3 pears cost 19 000; 2 apples and 4 pears cost 16 000

ANSWER Apple 2000, pear 3000

3. A boat covers 60 km down in 3 h and 60 km up in 5 h

ANSWER Boat 16, current 4

4. Two angles of a triangle differ by 30° and the third is 60°

ANSWER 45° and 75°

5. A father is 4 times his son and their ages add to 50

ANSWER Son 10, father 40

6. Why must a digit answer be whole and under 10?

ANSWER Digits are the symbols 0 to 9

7. Two numbers have sum 20 and one is 4 times the other

ANSWER 4 and 16

07 Homework

Homework — 5 tasks

Write both “let” sentences in words, and answer with both quantities.

- Two numbers add to 26 and differ by 8. Find them.
- 4 notebooks and 3 pens cost 26 000 sum; 2 notebooks and 1 pen cost 12 000. Find each price.
- The digits of a two-digit number add to 12 and the reversal is 18 larger. Find the number.
- A rectangle has perimeter 48 cm and its length is 6 cm more than its width. Find its sides.
- A boat does 22 km/h downstream and 14 km/h upstream. Find both speeds.